

1 Solution — GIAM 8.3

Problem 5

1.1 Problem

Cantor's theorem, applied to the set of all sets, leads to an interesting paradox. The power set of the set of all sets is a collection of sets, so it must be contained in the set of all sets. Discuss the paradox and determine a way of resolving it.

1.2 Stating the Paradox

Let

$$V = \text{"the set of all sets"} = \{x : x \text{ is a set}\}.$$

This V seems like it should make sense — for every set x , x has been “collected” into V . From naive intuition, V is just *the universe of set-theoretic discourse* viewed as a single object.

Two unavoidable facts now collide.

Fact 1 — Cantor's theorem. For any set X , $|X| < |\mathcal{P}(X)|$. Applied to V :

$$|V| < |\mathcal{P}(V)|.$$

Fact 2 — Inclusion. Every element of $\mathcal{P}(V)$ is a subset of V , and every subset of V is *itself a set*. So every element of $\mathcal{P}(V)$ is a set — hence an element of V . This says $\mathcal{P}(V) \subseteq V$, giving

$$|\mathcal{P}(V)| \leq |V|$$

(via the inclusion function $\mathcal{P}(V) \hookrightarrow V$, which is injective).

Combining the two facts:

$$|V| < |\mathcal{P}(V)| \leq |V| \implies |V| < |V|,$$

a flat contradiction. Cardinality is *not* strictly less than itself.

Cantor's Paradox: V Cannot Be a Set

Assume the "set of all sets" V exists. Two unavoidable facts collide.

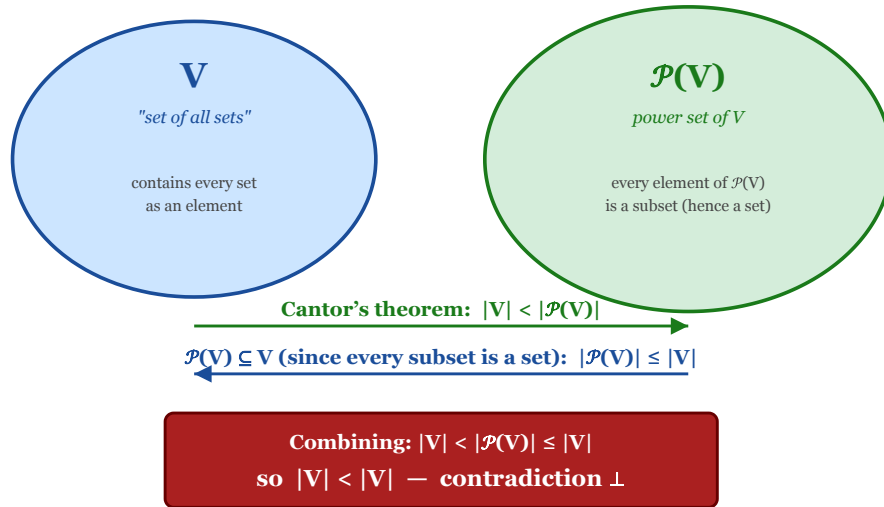


Figure 1.1: Two unavoidable consequences of assuming V is a set: Cantor's theorem ($|V| < |\mathcal{P}(V)|$) and the inclusion $\mathcal{P}(V) \subseteq V$ ($|\mathcal{P}(V)| \leq |V|$). Combined, they force $|V| < |V|$ — a contradiction. The assumption that V exists as a set must therefore fail.

1.3 How Did We Get Here?

The paradox traces back to a single tempting principle: *every property defines a set*. In Gottlob Frege's 1893 *Grundgesetze der Arithmetik*, this is the **comprehension principle**:

$$\{x : P(x)\} \text{ is a set, for any property } P.$$

Applied to $P(x) = "x \text{ is a set}"$, this principle directly conjures V into existence. Russell's paradox (1901) is the same principle applied to $P(x) = "x \notin x"$, and Cantor's paradox (informally known to Cantor by 1899) is what you get from V plus Cantor's theorem.

So the issue is not Cantor's theorem itself — that's a perfectly sound result about *any* set. The issue is the *premise* that V is a set in the first place. Drop that premise and the paradox evaporates.

1.4 The Resolution: V Is Not a Set

In modern set theory, the paradox is resolved by *axiomatically restricting* which collections count as sets. The naive comprehension principle is too generous; it allows collections “too big” to be sets. The fix is to start with a controlled list of axioms (Zermelo, 1908; Fraenkel and Skolem, 1920s) that lets sets exist only under carefully circumscribed conditions.

In **Zermelo–Fraenkel set theory with Choice (ZFC)** — the modern standard — comprehension is replaced by the *Axiom Schema of Separation*:

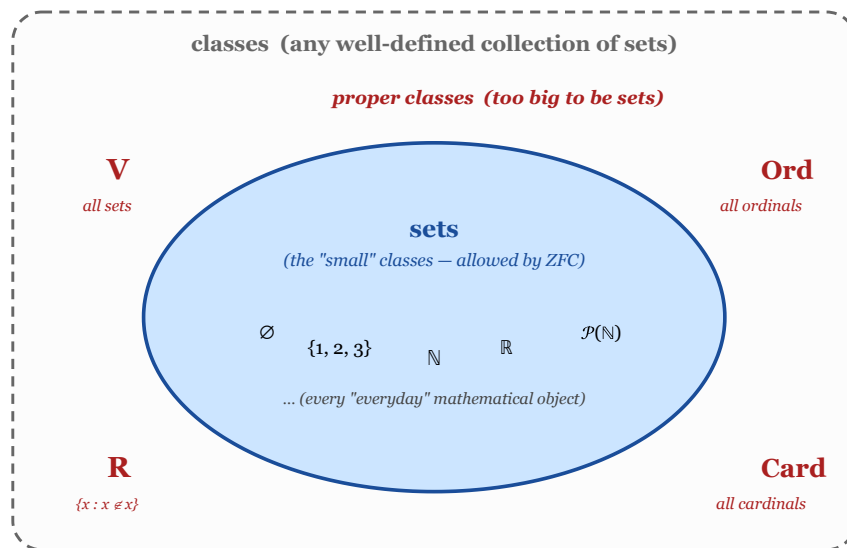
Given an existing set A and any property P , the collection $\{x \in A : P(x)\}$ is a set.

The crucial phrase is “*given an existing set A* ”: separation does not let you construct sets out of thin air; it only carves subsets out of pre-existing sets. There is no ZFC axiom granting “the collection of all sets exists.” So V simply *isn’t a set* in ZFC.

This dissolves Cantor’s paradox: we can still *talk about* the collection of all sets (it’s a meaningful concept), but we can’t apply Cantor’s theorem to it, because Cantor’s theorem is a statement about *sets* — and V isn’t one.

Resolution: V Is a Proper Class, Not a Set

ZFC restricts which collections count as “sets”; V is too big to be one.



*ZFC’s Separation axiom says $\{x \in A : P(x)\}$ is a set only when A already is.
There is no axiom granting “ V is a set,” and without that, Cantor’s paradox can’t even be stated.*

Figure 1.2: Modern set theory distinguishes ‘sets’ (the small classes that ZFC’s axioms allow) from ‘proper classes’ (collections too big to be sets, like V , Ord , Card , or Russell’s R). V is a proper class — meaningful as a concept but not an object you can substitute into Cantor’s theorem.

A collection like V that *isn’t* a set but is still well-defined is called a **proper class**. Examples (in addition to V):

- **Ord** — the collection of all ordinal numbers (Burali-Forti’s paradox shows this can’t be a set).
- **Card** — the collection of all cardinal numbers.
- $R = \{x : x \notin x\}$ — Russell’s collection.

These all share the property that *assuming they are sets leads to contradiction* — they live one rung above sets in the type hierarchy.

1.5 The Specific Mechanism — Separation in Detail

To make the resolution concrete, consider why naive comprehension fails but separation does not. Naive comprehension:

$$\{x : x \text{ is a set}\} \stackrel{?}{=} V \quad (\text{naive: yes; ZFC: not asserted}).$$

ZFC’s separation:

$$\{x \in A : x \text{ is a set}\} = A \quad (\text{when } A \text{ is already a set}).$$

To get V as a set via separation, you would need to start with some set A that *already contains all sets* — but that’s V itself, so the argument is circular. Without an a priori set “containing all sets,” separation cannot bootstrap V into existence. The other ZFC axioms (Pairing, Union, Power Set, Infinity, Replacement, Foundation, Choice) likewise build new sets from old, but none of them produces V .

So V is exactly the *unconstructible* collection in ZFC, and Cantor’s paradox is *not* the discovery of a contradiction in mathematics — it is the discovery of a contradiction in *one specific axiom* (naive comprehension) that was widely assumed before 1900.

1.6 Alternative Resolution: Treat Classes as Objects (NBG)

A different fix is **von Neumann–Bernays–Gödel set theory (NBG)**, developed in the \$ 1920 \$s–\$ 1940 \$s. NBG accepts classes as legitimate first-class citizens but distinguishes:

- **Sets:** classes that *can* be elements of other classes.
- **Proper classes:** classes that *cannot* be elements of other classes.

Then V is a proper class. The phrase " $\mathcal{P}(V)$ " doesn't even type-check, because the power-set operation only applies to sets — and V isn't one. Cantor's theorem, in NBG, is restricted to sets, so it never applies to V .

NBG and ZFC prove the same theorems about sets; NBG simply has a slightly richer vocabulary (it can quantify over proper classes directly, whereas in ZFC such statements have to be paraphrased into schemas). For everyday mathematics, the two systems are interchangeable.

A still more permissive approach is **Tarski–Grothendieck set theory** (or “ZFC with universes”), which posits an unbounded tower of *inaccessible cardinals* $U_1 \in U_2 \in \dots$, each U_i being a set that “looks like” the universe of sets, but the collection of *all* U_i is again a proper class. This is the framework used in some category-theoretic contexts where one wants to talk about “the category of all small categories” without circularity.

1.7 Remark — Cantor's Paradox vs. Russell's vs. Burali-Forti's

The three “size paradoxes” of the foundational crisis all share the diagonal-flip engine of Problem 4:

Paradox	Discovery	Diagonal collection that can't exist	Where Cantor's recipe applies
Burali-Forti	1897	Ord , the set of all ordinals	implicit
Cantor	1899	V , the set of all sets	\$

Russell	1901	$R = \{x : x \notin x\}$	Cantor's diagonal applied to V
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Table 1.1

All three are resolved the same way: deny that the offending collection is a set. The historical sequence is also illuminating — by the time Russell publicized his paradox in 1901, the broader class of “size-of-the-universe” paradoxes was already known to specialists. Russell’s was simply the cleanest example, requiring no setup beyond first-order logic and the comprehension principle.

1.8 Remark — Why This Was a Big Deal

Before 1900, Cantor’s theorem, Frege’s logicism, and the cumulative hierarchy of sets were all being developed in parallel, each implicitly assuming naive comprehension. The discovery that comprehension was inconsistent — first by Cantor himself (privately), then by Russell (publicly) — sent shockwaves through the foundations of mathematics:

- Frege’s 1893 *Grundgesetze* (volume 1) was already published. After Russell’s 1902 letter pointing out the paradox, Frege famously wrote in volume 2’s afterword: “*Hardly anything more unwelcome can befall a scientific writer than that one of the foundations of his edifice be shaken after the work is finished.*”
- Zermelo (1908), Fraenkel (1922), Skolem (1922), and others worked out axiom systems that proved consistent (relative to other accepted foundations) and powerful enough to develop the rest of mathematics.
- The eventual settled axiom system, ZFC, was in place by the late \$ 1920 \$s and remains the de facto foundation of mathematics today.

So Cantor’s paradox is not a curiosity — it’s a load-bearing entry in the historical narrative that produced modern axiomatic set theory.

1.9 Remark — Why You (Probably) Don’t Encounter This in Practice

For everyday mathematics — number theory, analysis, algebra, geometry, even most of category theory — you never need to consider whether “all sets” form a set. The collections you actually work with (groups, manifolds, topological spaces, sequences of reals,

...) are always sets by construction, built up from $\mathbb{N}$, $\mathbb{R}$, and the standard set-forming operations.

Cantor's paradox surfaces only when you try to take *unrestricted* unions, intersections, or images over “all of mathematics” — and even then, the modern habit is to confine yourself to a fixed *universe* U_i that is locally large enough for the problem. The paradox is a foundational fact rather than a working tool, and the resolution (ZFC's separation, or NBG's class/set distinction) is structured precisely so that *working mathematics* proceeds without disruption.

The principal upshot: *unrestricted comprehension is unsound; restricted comprehension via Separation is*. Every set you've ever used is built by restricted comprehension or one of the other ZFC operations, and the system as a whole has stood for almost a century with no contradiction discovered. Cantor's paradox marks the moment when set theory grew up.